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Instructions

This quiz will use the Coursera Lab Jupyter notebook "Lab to demonstrate some causes for Kalman-filter failure" from this lesson.

In another browser tab or window, open the lab. Run the lab and enter results from the lab as your responses to the final two questions in this practice quiz.

1. Which of the following conditions cause a Kalman filter to produce unreliable estimates? (Select all that apply.) 1 / 1 point

☒ If the Kalman filter uses an incorrect model of the dynamic system.

Correct
Yes. This is correct.

☒ If the process and/or sensor noises are nonzero-mean.

Correct
Yes. This is correct.

☒ If the process and/or sensor noises are autocorrelated.

Correct
Yes. This is correct.

☒ If the process and/or sensor noises are not Gaussian.

Correct
Yes. This is correct.

2. Run the code cells in the notebook from the section titled: "Implementing a Kalman filter with an incorrect dynamic model". What is the fraction of time that the position-estimate error is outside its 3σ error bounds? Enter your answer with two digits to the right of the decimal point. 1 / 1 point

Hint: You can compute this answer using the following line of code:

```
length(find(abs(xerr(1,:))>boundIMstore(1,:)))/nt
```

0.38961

Correct
Yes. That is correct.

3. Run the code cells in the notebook from the section titled: "Implementing a Kalman filter with biased (nonzero-mean) noises". What is the fraction of time that the velocity-estimate error is outside its 3σ error bounds? Enter your answer with three digits to the right of the decimal point. 1 / 1 point

0.95904

Correct
Yes. That is correct.